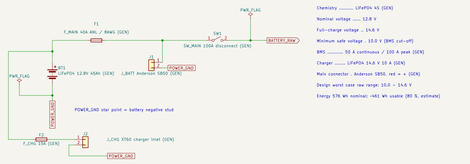
Battery and Charging

KIRO Wiring V2 | Battery_and_Charging

Battery, main fuse, disconnect, charger inlet and POWER_GND star point



BATTERY — GENERIC ASSUMPTION (replace with label data)

Chemistry ..... LiFePO4 4S (GEN)

Nominal voltage ..... 12.8 V

Full-charge voltage .. 14.6 V

Minimum safe voltage . 10.0 V (BMS cut-off)

BMS ..... 50 A continuous / 100 A peak (GEN)

Charger ..... LiFePO4 14.6 V 10 A (GEN)

Main connector . Anderson SB50, not = = (GEN)

Design worst case raw range: 10.0 – 14.6 V

Energy 576 Wh nominal; ~461 Wh usable (80 %, estimate)

J2/P2: charger joins on the battery side of SW_MAIN, so the robot can charge while switched off.

Reverse polarity: J_BATT must be keyed: compute branch adds an ideal-diode stage (Power_Distribution).

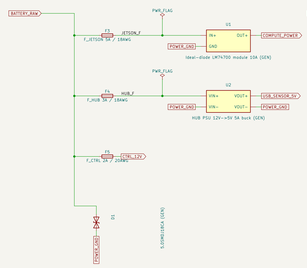
F_MAIN within 150 mm of BT1+. ~40 A above 25.6 A peak load, below 50 A BMS and 8 AWG rating.

Power Distribution

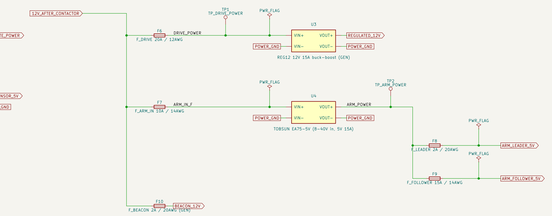
KIRO Wiring V2 | Power_Distribution

FB1 = unswitched fuse box (compute side) | FB2 = switched fuse box after contactor K1 (actuator side)

FB1 UNSWITCHED (not cut by E-stop)



FB2 SWITCHED (cut by E-stop via K1)



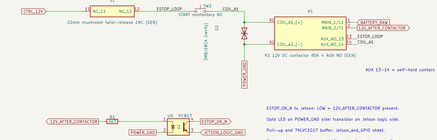
NOTES

1. Every branch fuse sits in FB1/FB2 at the start of its wire. Wire gauge shown in each fuse value: fuse <= wire and connector rating.
2. REG12 (REGULATED_12V) is required for lead-acid or LiFePO4: ST3215 limit 12.6 V, raw battery reaches 14.4–14.6 V. Omit only for a 3S Li-ion pack.
3. U1 reverse-polarity stage protects the Jetson branch. High-current branches rely on keyed J_BATT + F_MAIN.
4. TVS_BUS 18 V stand-off: above LiFePO4 14.6 V charge voltage (GEN — recheck if chemistry differs).
5. Pre-charge not fitted: generic REG12 input capacitance is small for an 80 A contactor. Re-check with the real REG12.
6. Arm 5 V rail: EA75-5V as in V1. Follower servo type on V2 (7.4 V or 12 V) is TBD — 5 V is safe for both.
7. HUB PSU stays on the unswitched side so cameras, LiDAR and adapter logic keep working after an E-stop.
8. #FLG = ERC power flags only (will enters through a passive fuse). Not physical parts.

Safety EStop and Contactor

KIRO Wiring V2 | Safety_EStop_and_Contactor

Hard-wired self-holding contactor loop. No software in the safety path.



ESTOP_OK_N to Jetson: LOW = 12V_AFTER_CONTACTOR present.

Opto LED on POWER_GND side: transistor on Jetson logic side.

Pull-up and 74LVC1G17 buffer: Jetson_and_GPIO sheet.

Broken wire → pull-up → HIGH → software treats as E-stop.

E-STOP BEHAVIOUR

Press mushroom → NC 11–12 opens → coil current stops → K1 opens → 12V_AFTER_CONTACTOR = 0 V (wheels, arms, light).

Broken loop wire or blown F_CTRL → same drop-out. Releasing the mushroom does not re-close K1 (AUX 13–14 open).

Restart: twist-release mushroom, then press START deliberately. COMPUTE_POWER is upstream of K1: Jetson stays up.

D2 bidirectional TVS clamps coil kick-back without the slow release of a plain flyback diode. Verify vs coil datasheet.

GENERIC: E-stop 1NC block 11–12; K1 12 V coil <= 1.5 A, main 80 A DC, AUX NO 13–14. Replace with real part data.

If K1 has no AUX NO: add a 12 V relay, coil parallel to K1 coil, and use its NO contact for the self-hold.

Jetson and GPIO

KIRO Wiring V2 | Jetson_and_GPIO

NVIDIA Jetson Orin Nano Super Developer Kit, reference carrier; J16 DC jack, J12 40-pin header, J6/J7 USB

J12 HEADER FACTS (NVIDIA carrier spec)

All J12 signals 3.3 V. Non-I2C pins go through TI TXB0108 level translators: drive only +/−20 uA.

External pull-ups/pull-downs on these pins must be > 50 kOhm. Never load a J12 pin directly.

Therefore every GPIO here meets a 74LVC1G17 buffer: (CMOS input, push-pull output).

Pins 29/31/7 are PROPOSED. Pinmux, boot state and state after cleanup: bench-verify (TBD).

J16: 9–20 V, 5.5 x 2.5 mm, centre positive, max 3.0 A. Raw battery 10–14.6 V is inside range.

Do NOT use hardware PWM for BEACON_HOLDOFF_PULSE: it must stop when software freezes.

USB PORT ASSIGNMENT

J6 port 1: Orbbec Gemini 335 (USB 3.2 Gen1, bundled cable, never via hub)

J6 port 2: powered USB hub upstream (LOAD + 3x CH343 servo adapters)

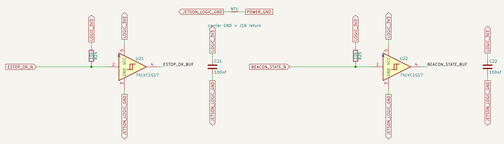
J7 port 1: Sujih wrist webcam. J7 port 2: spare. Each J6/J7 stack: 3 A VBUS total.

J15 GbE, J5 USB-C, J20/J21 CSI, J14 buttons: present on the carrier, not used in V2 (no-connect).



R21 1k GPIO -> buffer input (light sheet). R22/R23 100R buffer output -> TXB0108 B-side (overdrive).

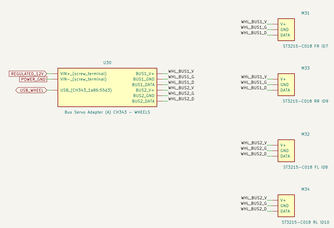
NT1: J12 GND and J16 DC return are the same carrier-board ground. No extra wire to POWER_GND.



Wheel Drive

KIRO Wiring V2 | Wheel_Drive

4 x Feetech/Waveshare ST3215-C018 smart serial servos on one CH343 bus adapter (no generic motor driver)



Chain 1 (BUS1): adapter -> FR ID7 -> RR ID9. Chain 2 (BUS2): adapter -> FL ID8 -> RL ID10. Max two servos per Molex 5264 contact (~3 A).

Supply: REGULATED_12V (ST3215 6–12.6 V; V1 Max_Voltage_Limit 140 V). Adapter powered via screw terminal, not its DC jack.

Current: idle ~1.2 A total, design 8 A, all four stalled 10.8 A (2.7 A each). Feed 14 AWG to screw terminal; 20 AWG servo leads.

Servo cable: 3-pin Molex 5264 (V+, GND, DATA). Physical pin order: verify against Waveshare ST3215 wiki before crimping (TBD).

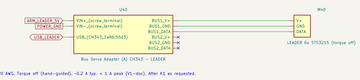
Data: half-duplex TTL 1 Mbps. USB_WHEEL via powered hub. Servo GND returns to the adapter, never through the USB cable shield.

E-stop: K1 removes REGULATED_12V -> servos unpowered, no holding torque, robot coasts. Speed feedback over bus = ROBOT_MOVING.

Leader Follower Arms

KIRO Wiring V2 | Leader_Follower_Arms

LeRobot leader + follower arms, each on its own CH343 bus adapter and its own fused 5 V branch



LEADER: F_LEADER 2 A, 20 AWG. Torque off (hand-guided). ~0.2 A typ, < 1 A peak (V1-doc). After K1 as requested.

FOLLOWER: F_FOLLOWER 15 A, 14 AWG. ~1–2 A typ, ~6 A design, ~9 A peak at 5 V (V1-doc, verify on V2).

Voltage: V1 follower = 7.4 V ST3215 (http 8.0 V) -> keep 5 V. A 12 V ST3215 variant would need a new rail (D2, TBD).

E-stop: K1 cuts EA75-5V input -> both arms limp. Follower falls under gravity: keep people clear of the arm envelope.

Bus 2 of each adapter unused (no-connect). USB to powered hub. Power return by its own wire to the star point.

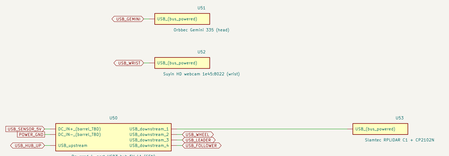


Adapter feed via screw terminal. Vendor nominal Input 9–12.6 V; V1 runs these adapters from 5 V successfully (V1-measured) — keep, verify.

USB Hubs Cameras and LiDAR

KIRO Wiring V2 | USB_Hubs_Cameras_and_LiDAR

Only the installed hub, two cameras and one LiDAR. No other sensors.



DEVICE TABLE

Gemini 335 | USB bus 5 V | 896 mA max, <6.5 W | USB 3.2 Gen1 | bundled cable | Jetson J6 port 1 direct

Wrist webcam | USB bus 5 V | needs 500 mA | USB 2.0 | camera cable | Jetson J7 port 1 direct

RPLUSA4 C1 | 5 V via USB | 230 mA typ, peak TBD | UART 460800 inside, USB outside (CP2102N) | hub D51

USB hub | DC in 5 V 4 A barrel 5.5 x 2.1 mm, centre positive (GEN — check against real hub label)

hub barrel: if the real hub is 5.5x2.35 or 5.5x2.5 mm, change the plug: never trust "5 mm" alone.

Shielding: use shielded USB cables: route away from servo and 12 V harnesses. No motor current returns via USB.

LiDAR start-up (V1): plug sensor cable into adapter with USB unplugged. then connect USB. Motor spin-up after host opens port.

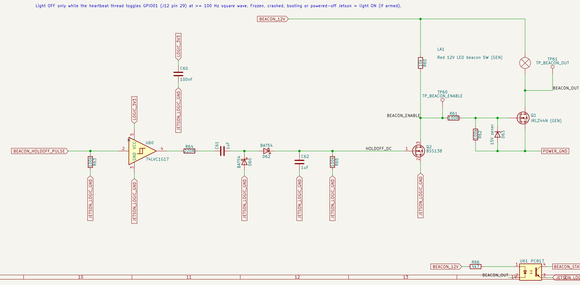
Red Motion Warning Light

KIRO Wiring V2 | Red_Motion_Warning_Light

Fail-to-warning: light is ON whenever K1 is closed, unless the Jetson actively holds it OFF with a live pulse train

SOFTWARE MEANING: MOTION_WARNING_REQUEST active = stop toggling BEACON_HOLDOFF_PULSE = light ON

Light OFF only while the heartbeat thread toggles GPIO01 (J12 pin 29) at >= 100 Hz square wave. Frozen, crashed, booting or powered-off Jetson = light ON (if armed).



Connectors and Test Points

KIRO Wiring V2 | Connectors_and_Test_Points

Test-point panel on every cross-sheet rail, plus the connector schedule

CONNECTOR SCHEDULE (rating must exceed fuse; pinout verified before crimping)

J_BATT (J1) Anderson SB50, keyed, 50 A ..... GEN

J_CHG (J2) XT60 charger inlet ..... GEN

J16 Jetson 5.5 x 2.5 mm barrel, centre +, 3.5 A max - NVIDIA spec

Adapter VIN screw terminal (not the 5.5 x 2.1 jack) — Waveshare

Servo bus Molex 5264 3-pin, use factory servo leads

Hub DC in 5.5 x 2.1 mm, centre +, 5 V 4 A ..... GEN

J_BEACON (J60) 2-pin 5 A, pin 1 = lamp + ... GEN

ESTOP / START panel wiring, 20 AWG, ferrules ..... GEN

K1 main M6 ring lugs, 80 A DC contact ..... GEN

Signal harness J12 pins 1/6/29/31/7 to buffer board, 2.54 mm Dupont, strain-relieved

TEST-POINT USE

Measure all rails against TP_POWER_GND at the star point. Local TPs:

TP1 DRIVE_POWER, TP2 ARM_POWER (Power_Distribution); TP60 BEACON_ENABLE.

TP61 BEACON_OUT (Red_Motion_Warning_Light). Strain relief on every harness exit.

Fleshened from hierarchical sheets for single-file delivery

Genie Enrtech

Sheet: /

File: kiro_wiring_v2.kicad_sch

Title: KIRO Wiring V2 (flat)

Size: A0	Date: 2026-05-12
KiCad E.D.A. 10.0.1	Rev: flat-1
	Id: 1/1